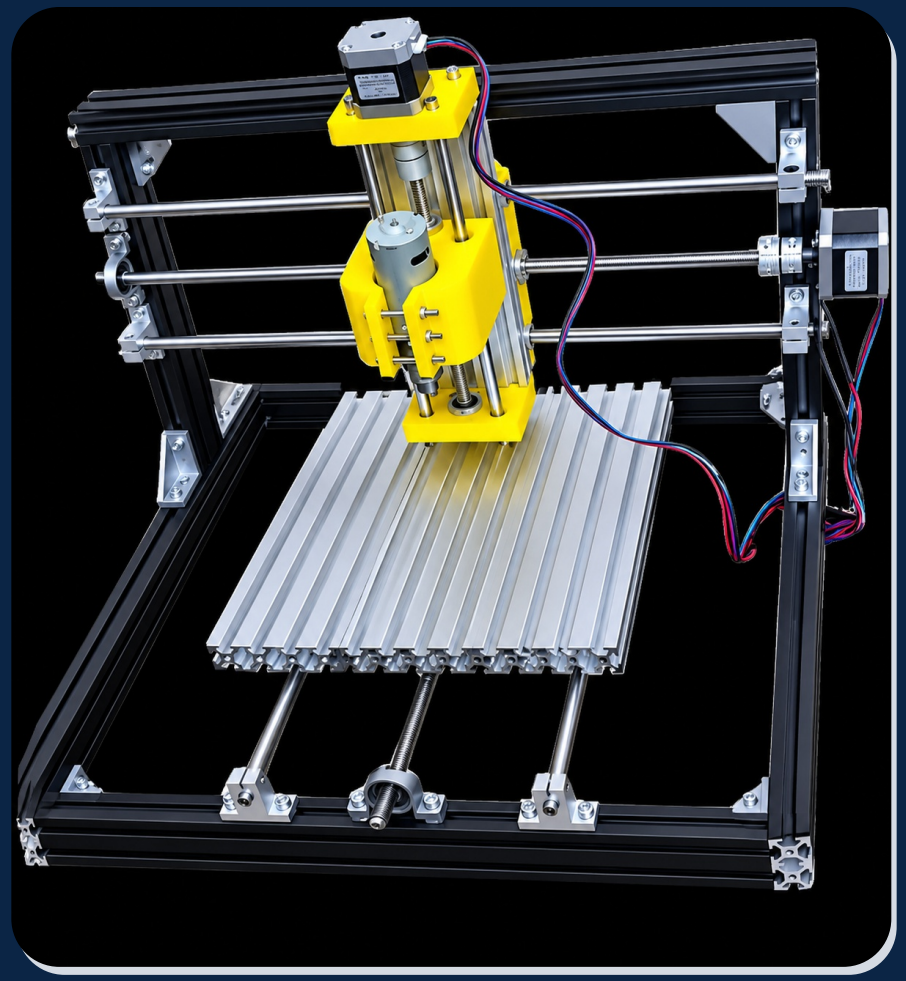
PROJECT PRESENTATION

Design and Implementation of a CNC Machine for PCB Manufacturing

A low-cost, compact CNC machine for precise PCB drilling and milling



Project Definition & Objectives

PROJECT DEFINITION

Printed Circuit Boards (PCBs) are fundamental components in all electronic devices. In this project, a simple CNC (Computer Numerical Control) machine was designed and implemented for the precise drilling and manufacturing of PCBs using low-cost tools and techniques. The aim is to reduce reliance on traditional chemical etching and provide a practical, rapid solution for students and developers.

Project Objectives

1. Design a small-sized CNC machine dedicated to PCB manufacturing.
2. Implement a precise motion control system for the machine's axes.
3. Use CAM software to convert PCB designs into G-code.
4. Produce functional, usable PCB boards for real-world electronic projects.

Project Components

Mechanical

- Stepper motors
- Linear rails for X, Y and Z axes
- Aluminum frame

Electronic

- Arduino UNO / CNC Shield
- Motor driver (A4988 or DRV8825)
- Power supply (12 V)
- End stops (limit switches)
- Drilling machine (mini drill or small spindle)

Programmatic

- PCB design software such as Circuit Wizard
- Convert the design to G-code with FlatCAM
- Operate the machine with Universal G-code Sender or bCNC

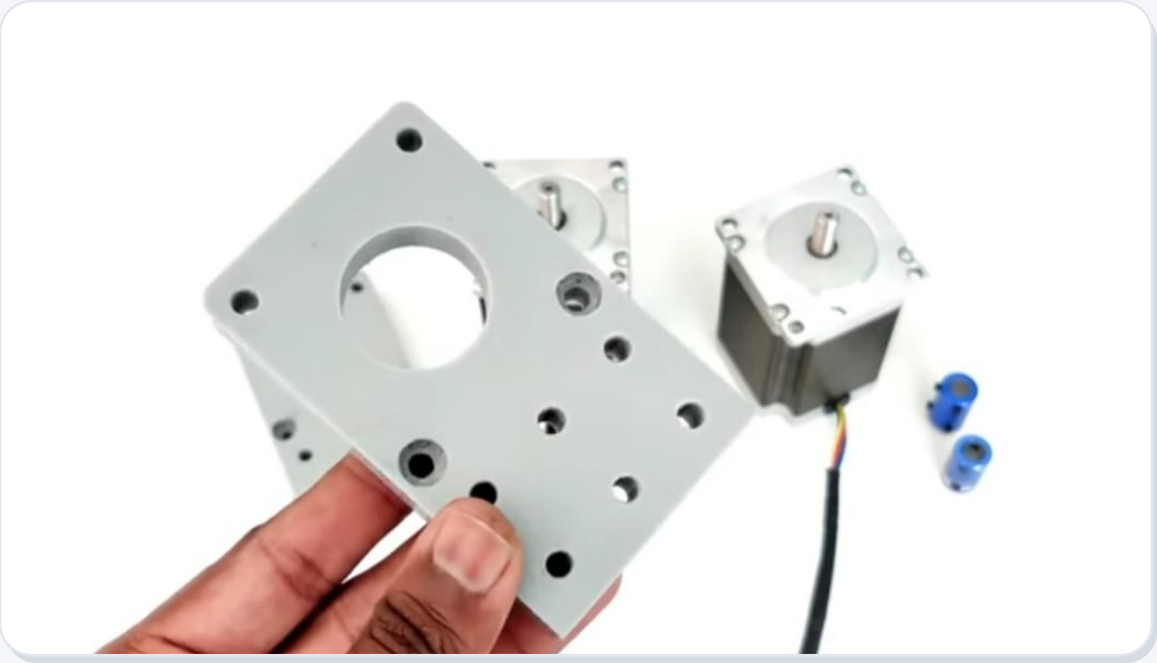
1 Stepper Motor Mounting Bracket

FUNCTION

Securely mounts the stepper motor onto the structure of a project or device, such as 3D printers, CNC machines or robots.

SPECIFICATIONS & MOUNTING HOLES

- The large circular hole in the center lets the motor shaft pass through.
- The smaller holes match the motor mounting screws and also secure the bracket to the project structure.



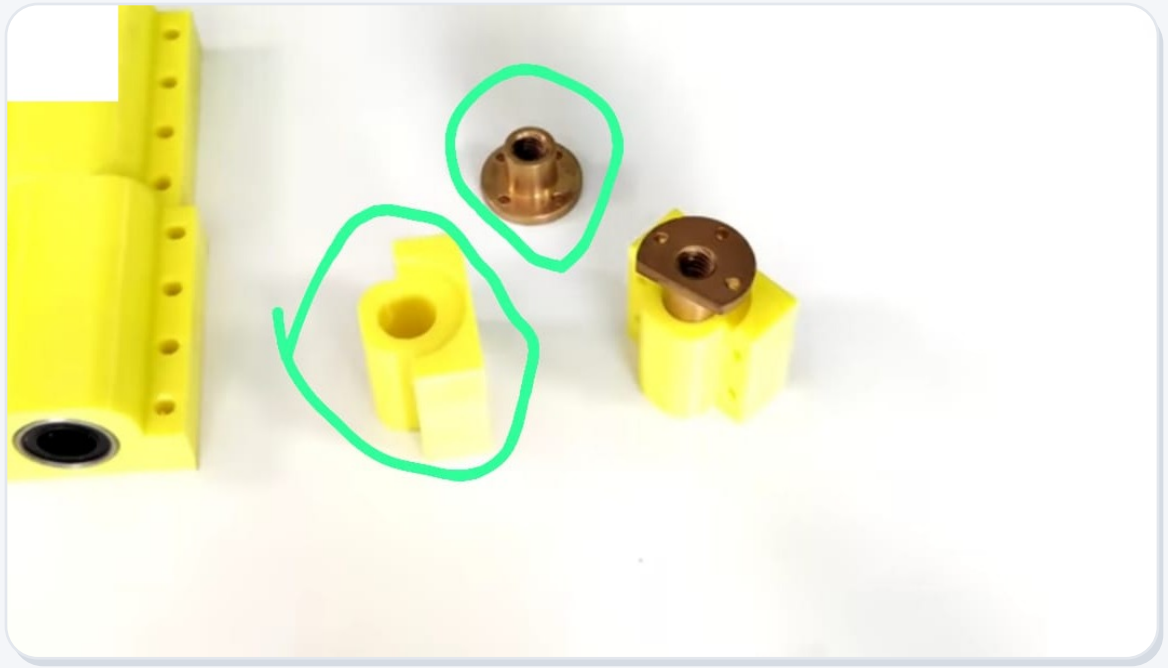
2 Screw Nut Brass Lead & Plastic Nut Holder

SCREW NUT BRASS LEAD

- Used with the lead screw to convert rotational motion into linear motion.
- Made of brass to reduce friction and noise.
- Common in axis systems of 3D printers, CNC machines and automation projects.

PLASTIC NUT HOLDER

- Secures the brass lead screw nut onto the device structure.
- The design includes a space for mounting the nut and adjusting its orientation.
- Side holes: used to fix the nut to the holder with screws.



3 DC Motor 12 V

FUNCTION AND USE

- Miniature drilling tools (such as a mini drill or Dremel).
- Simple CNC projects or robots.
- DIY projects that need high-speed rotation power.

COMMON SPECIFICATIONS

- Voltage: 12 V
- Speed: can reach 10,000 rpm or more, depending on the model.



4 Lead Screw

FUNCTION

Converts rotational movement (usually from the motor) into precise linear movement.

HOW IT WORKS

- A specially designed nut is fixed onto the screw.
- When the screw rotates, the nut moves along its length, or vice versa.
- In this way, a mechanical part can be moved back and forth with high precision.



5 Motor Mounting Plate

FUNCTION

Securely mounts the stepper motor onto the main frame of the machine, such as CNC machines or 3D printers.

Hole description

1 Large central hole

Allows the motor shaft to pass through.

2 Four surrounding holes

Used to fix the motor to the plate with screws.

3 Other scattered holes

Used to mount the plate itself onto the aluminum frame.

6 Spindle Holder

FUNCTION

Secures the spindle motor in place on the movement axis (usually the Z-axis). The spindle rotates at high speed and holds the cutting tool or drill bit.

FEATURES

- **Large circular opening:** the base of the spindle is inserted here.
- **Bottom holes:** mount the holder onto the moving part of the Z-axis.
- **Upper side holes:** tightly secure the spindle around its circumference.



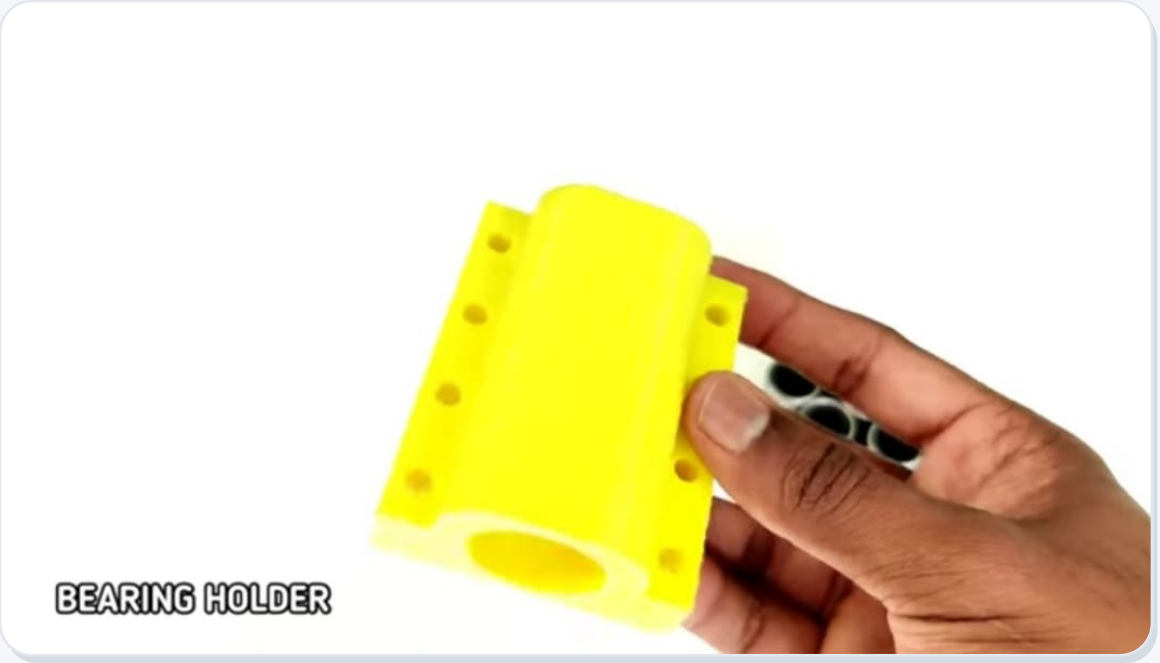
IN USE

When the Z-axis moves up and down, this holder moves along with the spindle, allowing the machine to cut or engrave materials according to the CNC design.

7 Bearing Holder

FUNCTION

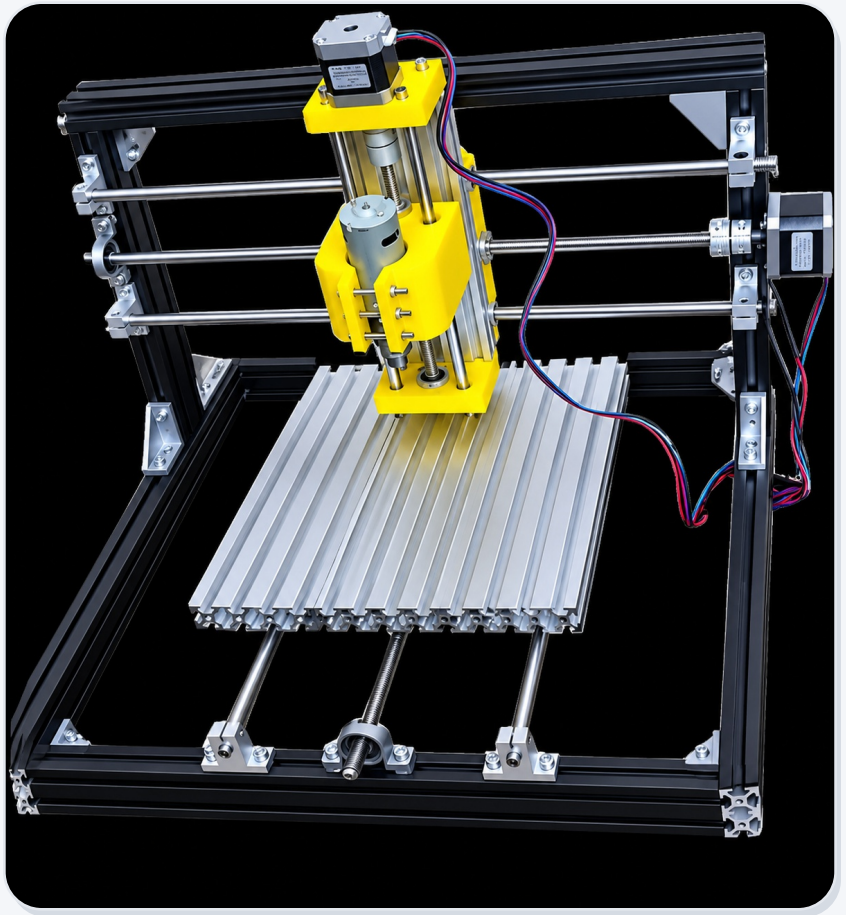
- Holds the bearing securely in place within any mechanical system.
- The bearing itself reduces friction between moving parts, such as rotating shafts.
- The holder fixes the bearing onto a stable base (like the machine chassis) and keeps it stable for efficient performance.

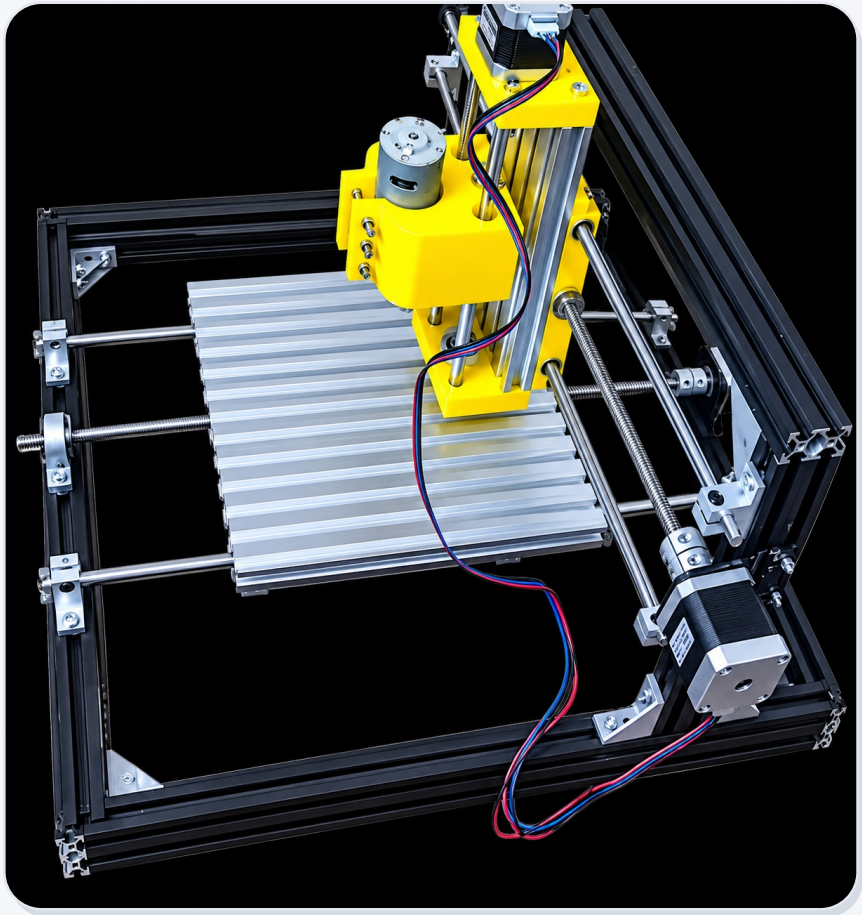


Machine Operation Steps

1. Design the electronic circuit using software like Circuit Wizard.
2. Export the design in Gerber format.
3. Use FlatCAM to convert the Gerber files into G-code.
4. Load the G-code into the control software (such as UGS or bCNC).
5. Mount the copper board onto the CNC table.
6. Start the automatic milling process.
7. Inspect the result and make adjustments if necessary.

The Final Machine





The assembled CNC machine for PCB drilling and milling